

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development MERN Practical

M2 Practical-8

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Class:TYIT	Batch:1
Date of Assignment: 30-9-2026	Date/Time of Submission: 30-9-2026

a. Create a simple REST API using Express

Code:-

[server.js](#)

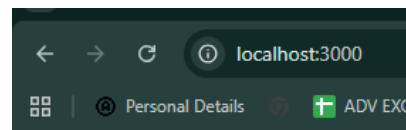
```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/CollegeD
B")
.then(() => {
  console.log("MongoDB Connected Successfully");
})
.catch((error) => {
  console.log("MongoDB Connection Failed");
  console.log(error);
});
// Simple REST API
app.get("/", (req, res) => {
  res.send("Student Management API is Running");
});
app.listen(PORT, () => {
  console.log(`Server is running on
http://localhost:${PORT}`);
});
```

Output:-

On terminal

```
PS C:\Users\itlab\Desktop\TYIT Aashu MERN\practical 8> node server.js
Server is running on http://localhost:3000
MongoDB Connected Successfully
```

On browser



Student Management API is Running

b. Implement a GET API to retrieve data.

Code:-

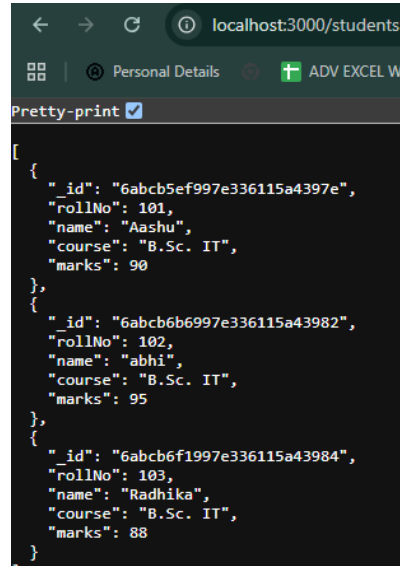
server.js:-

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/CollegeD
B")
.then(() => {
  console.log("MongoDB Connected Successfully");
})
.catch((error) => {
  console.log("MongoDB Connection Failed");
  console.log(error);
});
// Simple REST API
app.get("/", (req, res) => {
  res.send("Student Management API is Running");
});
// Get all students
app.get("/students", async (req, res) => {
  try {
    const students = await mongoose.connection.db
.collection("Students")
.find()
.toArray();
    res.json(students);
  }
  catch (error) {
    res.status(500).json({
      message: "Error fetching students"
    });
  }
});
// Start server
```

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```
app.listen(PORT, () => {
  console.log(`Server is running on
http://localhost:${PORT}`);
});
```

Output:-



c. Fetch data from the API in a React application.

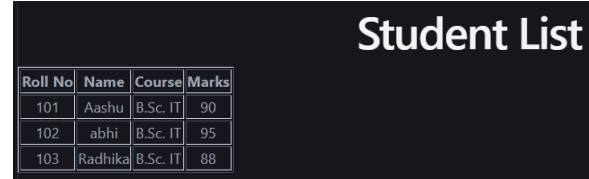
Code:-

```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
  const [students, setStudents] = useState([]);
  useEffect(() => {
    axios.get("http://localhost:3000/students")
      .then((response) => {setStudents(response.data);})
      .catch((error) => {console.log(error);}); }, []);
  return (
    <div>
      <h1>Student List</h1>
      <table border="1">
        <thead>
          <tr>
            <th>Roll No</th>
            <th>Name</th>
            <th>Course</th>
            <th>Marks</th>
          </tr>
        </thead>
        <tbody>
          {students.map((student) => (
            <tr key={student._id}>
              <td>{student.rollNo}</td>
              <td>{student.name}</td>
              <td>{student.course}</td>
              <td>{student.marks}</td>
            </tr>
          ))}
        </tbody>
      </table>
    </div>
  );
}
```

```

</tbody>
</table>
</div>
);
}
export default App;
Output:-

```



d. Send data from React to the server using a POST Request.

e. Store data in MongoDB.

f. Display stored data in React page.

server.js

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose
  .connect("mongodb://127.0.0.1:27017/CollegeDB")
  .then(() => {
    console.log("MongoDB Connected Successfully");
  })
  .catch((error) => {
    console.log("MongoDB Connection Failed");
    console.log(error);
  });
// Simple REST API
app.get("/", (req, res) => {
  res.send("Student Management API is Running");
});
// Get all students
app.get("/students", async (req, res) => {
  try {
    const students = await mongoose.connection.db
      .collection("Students")
      .find()
      .toArray();

    res.json(students);
  } catch (error) {
    res.status(500).json({
      message: "Error fetching students",
    });
  }
});
}
```

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```
});  
// Add a new student  
app.post("/students", async (req, res) => {  
  try {  
    const student = {  
      rollNo: req.body.rollNo,  
      name: req.body.name,  
      course: req.body.course,  
      marks: req.body.marks,  
    };  
    await mongoose.connection.db  
      .collection("Students")  
      .insertOne(student);  
  
    res.json({  
      message: "Student added successfully",  
    });  
  } catch (error) {  
    res.status(500).json({  
      message: "Error adding student",  
    });  
  }  
});  
// Start server  
app.listen(PORT, () => {  
  console.log(`Server is running on  
http://localhost:${PORT}`);  
});  
App.jsx  
import { useEffect, useState } from "react";  
import axios from "axios";  
function App() {  
  const [students, setStudents] = useState([]);  
  const [rollNo, setRollNo] = useState("");  
  const [name, setName] = useState("");  
  const [course, setCourse] = useState("");  
  const [marks, setMarks] = useState("");  
  // Get students  
  const getStudents = () => {  
    axios  
      .get("http://localhost:3000/students")  
      .then((response) => {  
        setStudents(response.data);  
      })  
      .catch((error) => {  
        console.log(error);  
      });  
  };  
  // Add student  
  const addStudent = () => {  
    const newStudent = {
```

```
      rollNo: Number(rollNo),  
      name: name,  
      course: course,  
      marks: Number(marks),  
    };  
    axios  
      .post("http://localhost:3000/students", newStudent)  
      .then((response) => {  
        alert(response.data.message);  
        setRollNo("");  
        setName("");  
        setCourse("");  
        setMarks("");  
        getStudents();  
      })  
      .catch((error) => {  
        console.log(error);  
      });  
  };  
  return (  
    <div>  
      <h1>Student Management System</h1>  
      <h2>Add Student</h2>  
      <input  
        type="number"  
        placeholder="Roll No"  
        value={rollNo}  
        onChange={(e) => setRollNo(e.target.value)}  
      />  
  
      <br />  
      <br />  
      <input  
        type="text"  
        placeholder="Student Name"  
        value={name}  
        onChange={(e) => setName(e.target.value)}  
      />  
  
      <br />  
      <br />  
      <input  
        type="text"  
        placeholder="Course"  
        value={course}  
        onChange={(e) => setCourse(e.target.value)}  
      />  
  
      <br />  
      <br />  
      <input  
        type="number"  
        placeholder="Marks"  
        value={marks}  
        onChange={(e) => setMarks(e.target.value)}  
      />  
    )  
  );  
}
```

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```
<br />
<br />
<button onClick={addStudent}>Add Student</button>
<hr />
<h2>Student List</h2>
<table border="1">
  <thead>
    <tr>
      <th>Roll No</th>
      <th>Name</th>
      <th>Course</th>
      <th>Marks</th>
    </tr>
  </thead>
  <tbody>
    {students.map((student) => (
      <tr key={student._id}>
        <td>{student.rollNo}</td>
        <td>{student.name}</td>
        <td>{student.course}</td>
        <td>{student.marks}</td>
      </tr>
    ))}
  </tbody>
</table>
</div>
);
}
export default App;
```

localhost:5173 says

Student added successfully

OK

Roll No	Name	Course	Marks
101	Aashu	B.Sc. IT	90
102	abhi	B.Sc. IT	95
103	Radhika	B.Sc. IT	88
104	riya	cs	55